MIL-E-17624B(SH)
11 August 1987
SUPERSEDING
MIL-E-17624A(SHIPS)
15 June 1960
(See 6.5)

MILITARY SPECIFICATION

EXPANDER, TUBE, ROLLER, BOILER TUBE

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers boiler tube expanders of the self-feeding parallel rolling type for use in expanding carbon steel and alloy steel tubes in all types of boilers.

1.2 Classification. Tube expanders shall be of the following types and classes, as specified (see 6.2):

Type I - Plain expander
  Class 1 - General application
  Class 2 - Economizer
  Class 3 - Superheater
  Class 4 - Air heater
Type II - Long reach expander
Type III - Reverse expander

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441
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SPECIFICATIONS

FEDERAL

PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Government drawings. The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

5000-S5101-1385500 - Expanders, Tube, Boiler, Assemblies and Details, Dimensions for.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Cages. Cages shall be made of heat-treated alloy steel, hardness 46-50 Rockwell C.

3.2.2 Mandrels and rolls. Mandrels and rolls shall be made of alloy tool steel, tempered and hardened to withstand the shocks and wear of regular usage, hardness 60-64 Rockwell C. Shanks on mandrels for expanders 2 inches and larger shall have hardness 46-48 Rockwell C. Mandrels and rolls shall be free from all bends. No decarbonization will be permitted on the rolls and mandrels.

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3.2.3 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Construction. Expanders shall consist of mandrels, cage, and three rows of rolls. Unless otherwise specified in the contract or order (see 6.2), expanders shall be right-hand feed. Expanders shall be configured so that the minimum diameter to which the tool expands will be reached before rolls bind.

3.3.1 Expanders, rolls and mandrels shall be furnished in the sizes shown on Drawing 5000-S5101-1385500. Unless otherwise specified (see 6.2), complete expanders shall be furnished with rolls and set of drum mandrels as specified on Drawing 5000-S5101-1385500. Each expander shall be supplied with an extra set of rolls in order to permit substitution of a long roll for the short and flare roll when rerolling tubes.

3.3.2 Cage. Cages shall be configured so that rolls cannot fall in or out when mandrel is withdrawn. Tolerance on the diameter of the cages shall be as follows:

(a) For cages for 1 to 1-1/2 inch expanders inclusive - plus or minus 0.003 inch.
(b) For cages for 2 inch expanders and larger - plus or minus 0.005 inch.

Cages shall conform to the dimensions shown on Drawing 5000-S5101-1385500 as required for each type and class of expander.

3.3.2.1 Cage slots. Slots shall be 120 degrees apart and configured to give a feed angle of 2 degrees minus 1/4 degree plus 0 degree for 1 to 1-1/2 inch expanders, inclusive, and 1-3/4 degrees minus 1/4 degree plus 0 degree for 2 inch expanders and larger. Tolerance on the length of roll slots shall be plus or minus 0.005 inch with a nominal clearance of 1/64 inch over roll length. Tolerance on width of roll slots shall be as follows:

(a) For 1 to 1-1/2 inch expanders inclusive - nominal clearance of 0.004 inch over major diameter of roll with a tolerance of plus 0.005 inch, minus 0.002 inch.
(b) For 2 inch expanders and larger - nominal clearance of 0.008 inch with a tolerance of plus 0.006 inch, minus 0.004 inch.

Slots shall be so configured that front end of all rolls are in the same plane.

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3.3.3 Rolls. Rolls shall have a taper of 1/64 inch per inch total taper over effective length of roll to produce parallel expansion of the tube. Effective length of roll shall be 3/8 inch beyond tube seat specified (see 6.2). Rolls shall be ground or polished with tool marks removed. They shall be rounded at the entrance end a sufficient amount to prevent flat end of roll from jamming into unrolled portion of the tube when expander is being used at the maximum expansion. Rolls shall be configured so that the short roll and flare roll may be removed and replaced by a long roll of the same set when rerolling of tubes is required without flaring. This does not apply to long reach expanders.

3.3.3.1 Tolerances on the diameter of rolls shall be as follows:

(a) For 1 to 1-1/2 inch expanders inclusive - plus 0.000, minus 0.003 inch.
(b) For 2 inch expanders and larger - plus 0.000 minus 0.005 inch.

3.3.3.2 Tolerances on the length of rolls shall be as follows:

(a) For all sizes of expanders - plus 0.000, minus 0.005 inch.

3.3.4 Mandrels. Mandrels shall have a taper of 1/32 inch per inch of mandrel length. Mandrels shall be supplied in four lengths, and shall be conconstructed for hand or power operation. Mandrels for the 1 inch up to, but not including, the 3-inch expanders shall have a 1/2 inch square shank. Mandrels for expanders for tubes 3 inches and larger shall have a 1 inch square shank. Tapered section of mandrels shall be ground or polished with toolmarks removed. Tolerances on diameters shall be plus or minus 0.004 inch. The square shank of each mandrel for expanders 2 inches and larger shall be drawn back to a Rockwell C hardness of 46-48. Mandrels may be supplied with a 1/2 inch straight section adjacent to the shank, if the manufacturer so desires. In the event mandrels are so supplied, major diameter of mandrel as specified on Drawing 5000-S5101-1385500 shall be decreased by 0.016 inch.

3.3.5 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.4 Type I, plain expanders. Plain expanders shall consist of mandrels, cage, two tapered long rolls, one tapered short roll, and one flare roll.

3.4.1 Class 1, general application and class 4 airheater. Cages for expanders for tube sizes up to, but not including 3 inches shall be configured to enter a handhole 3-3/8 by 3-3/4 inches. Cages for expanders for tube sizes 3 inches and larger shall be configured to enter a handhole 3-25/32 by 4-23/32 inches.

3.4.2 Class 2, economizer and class 3, superheater. Cages shall be configured to enter a handhole 2 inches in diameter.

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3.5 Type II, long reach expander. Long reach expanders shall consist of mandrels, cage, three tapered long rolls, and a steel thrust collar attached to the case to prevent its loss. Cages shall be constructed to expand tube seats 2-1/2 inches and larger in 2 operations. Long reach expanders shall expand the outer (fireside) end of seat. The inner (water side) end of tube shall be expanded with the appropriate type I, class 1 expander for 1-3/4 inches seat for the tube size required.

3.6 Type III, reverse expander. Reverse expanders shall consist of mandrels, cage, thrust collar, 2 tapered long rolls, one tapered short roll, and one flare roll. Flare roll shall be in opposite end of cage as compared to type I expanders. Expander shall be adjustable to expand inaccessible end of nipples 2-1/2 to 5 inches long. Reverse expanders shall be used in conjunction with appropriate plain expander for tube size and seat required in nipples where further seat is inaccessible.

3.7 Marking.

3.7.1 Expanders. Each expander cage shall be marked with the manufacturer's name and NAVSEA "E" number, as it appears on Drawing 5000-S5101-1385500, size of tube and tube wall thickness for which used, and tube seat length for which used.

3.7.2 Rolls. The designation as shown on Drawing 5000-S5101-1385500 shall be stamped on a linen tag and attached to the container in which rolls are supplied. The nominal tube seat length for which used, manufacturer's name, contract number, and item number shall also be included. Each set shall be individually wrapped and a linen tag bearing the same information as that on the container wrapped with set. Rolls will be purchased in sets for a complete expander.

3.7.3 Mandrel designation as shown on Drawing 5000-S5101-1385500 shall be stamped on the shank of each mandrel. In addition, manufacturer's name or mark shall be stamped on the shank.

3.7.4 In addition to 3.7.1, 3.7.2 and 3.7.3, each complete expander shall have a linen tag attached to it containing the following information:

(1) Manufacturer's name
(2) Contract number and item number
(3) Standard Navy stock number

3.8 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absense of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Prior to production, expander shall be examined and tested as specified in 4.6. During testing under 4.7.2, the tools shall be operated by the usual qualified personnel during normal maintenance tube expanding operations on boilers. After initial approval for satisfactory operation, first article inspection shall be repeated.

4.3.1 Samples for first article inspection. The contractor shall submit four complete units with drum mandrels for 1-3/4 inch tube sheet thickness covering each type expander for which he desires approval.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination specified in 4.6 and test specified in 4.7.1.

4.5 Sampling for quality conformance inspection.

4.5.1 Inspection lot. All assembled expanders, rolls, or mandrels of the same size offered for delivery at one time shall be considered an inspection lot.

4.5.2 Sampling for visual and dimensional examination. Sample rolls, mandrels, and assembled expanders shall be selected from each lot, at random, in accordance with MIL-STD-105 at inspection level II for the examination of 4.6. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.5.3 Sampling for hardness test. Sample rolls, mandrels, and assembled expanders shall be selected in accordance with MIL-STD-105 at inspection level I for the test specified in 4.7.1.

4.6 Visual and dimensional examination. Each of the assembled expanders selected in accordance with 4.5.2 shall be examined to determine conformance to this specification regarding dimensions, fits, finishes, security of assembly and freedom of movement of the parts. Each of the sample rolls and mandrels selected in accordance with 4.5.2 shall be examined to determine conformance to this specification regarding dimensions and finishes. Any sample roll, mandrel or assembly which does not conform to this specification shall be rejected, and if the number of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Examination shall be conducted as specified in table I.

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TABLE I. Classification of defects.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Expander assembly not complete; component parts missing.
102	Evidence of use of unauthorized material.
103	Expander not type and class specified.
104	Expanders, rolls and mandrels not of size specified.
105	Expander not right-hand feed (as required).
106	Cage diameter not within specified tolerance.
107	Cage dimensions and configuration not as specified.
108	Evidence that rolls fall out or in when mandrel removed.
109	Cage slots not angularly spaced as required; feed angle not within specified tolerance.
110	Length and width of roll slots not within specified tolerance; front end of all rolls not in same plane.
111	Taper on rolls over effective length nonconforming; not 1/64 inch per inch; length of roll nonconforming.
112	Front end of rolls not rounded.
113	Short roll, flare roll, and long rolls not interchangeable (types I and III only).
114	Length and diameter not within tolerance limits.
115	Mandrel taper nonconforming; not 1/32 inch per inch.
116	Mandrels not of specified number and length; mandrel shank not square; shank size not 1/2 or 1 inch.
117	Mandrel diameter not within tolerance limits.
118	Extra set of rolls not provided.
119	Tapered section of mandrels not ground or polished; tool marks not removed.
120	Thrust collar not attached to long reach expander (type II only).
121	Reverse expander, type III flare roll not in opposite end of cage as compared to plain expander.
122	Expander cage not marked with manufacturer's name and NAVSEA "E" number; not marked with size of tube, tube wall thickness and tube seat length for which used; markings illegible.
123	Mandrel designation and manufacturer's name not stamped on shank of mandrel; not legible.
124	Burrs, sharp corners not removed.
125	Mandrel (or roll) not free of bends.
126	Expander not left-hand feed (if specified).
Minor:	
201	Marking tag for rolls container not supplied; data missing, incomplete, not legible.
202	Sets of rolls not individually wrapped; linen tag not included.
203	Expander cage stamped marking not correct or not legible.
204	Mandrel stamped designation not correct or not legible.

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4.7 Test procedures.

4.7.1 Hardness test. Cages shall be tested for hardness specified in 3.2.1. Mandrels and rolls shall be tested for hardness specified in 3.2.2. Shanks on mandrels for expanders 2 inches and larger shall be tested for hardness specified in 3.2.2. Rolls and mandrels shall be tested for hardness on a part of the roll or mandrel where the mark will not interfere with their use. On the rolls the hardness test shall be performed on the rounded nose. On the mandrels the hardness test shall be performed on the last 1/2 inch of the tapered section, or if the alternate method of manufacture is used (see 3.3.4) on the 1/2 inch straight portion of the mandrel.

4.7.2 Operation test. The samples submitted as specified in 4.3.1 shall be tested for satisfactory operation. Satisfactory operation shall be construed to mean that at least two of the samples shall expand each of the following number of tube ends:

(a) 1 inch to 2 inch expanders, inclusive - 200 tube ends.
(b) 3 inches and larger - 50 tube ends.

The samples shall be considered to have failed the test if any of the following criteria are not met. Evidence that expander does not expand inaccessible end of nipples (type III only). Evidence of lack of freedom of movement of parts. Expander fails to expand specified test number of tube ends. Cage will not pass through specified size hand holes (type I only).

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging, packing and marking. Preservation-packaging, packing and marking shall be in accordance with MIL-P-116 and the following.

5.1.1 Tube expanders. Tube expanders shall be preserved and packaged level A or C; packed level A, B or C and marked as specified in accordance with PPP-P-40 (see 6.2).

5.1.2 Rolls and mandrels. Rolls and mandrels, when furnished separately, shall be preserved and packaged as specified for tube expanders. Shipping containers for level A or B packing shall be limited to 200 pounds gross weight for wood or wood-cleated boxes.

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6. NOTES

6.1 Intended use. The tube expanders are intended for use in the belling and expanding of boiler tubes in drums and headers of steam boilers. The expanders may be used with either hand- or power-operation and coupled with accessories, such as right-angle gear drivers, universal joints, extensions, and parallel gear drives as the situation requires.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) If expanders are other than right-hand feed (see 3.3).
(d) If expanders are not furnished with rolls and drum mandrels (see 3.3.1).
(e) Size of tube expander, rolls or mandrels required; also thickness of tube sheet in which tube is to be expanded to nearest larger 1/4 inch (see 3.3.3 and 3.4).
(f) Mark designation as shown on Drawing 5000-S5101-1385500, (see 3.7).
(g) Level of packaging and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Cage
Carbon
Mandrel
Steel

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-17624B(SH)	EXPANDER, TUBE, ROLLER, BOILER TUBE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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